
Part 2: Market Power Practice Problems

Econ 502: Advanced Microeconomics

1. Monopoly Pricing

A monopolist faces the inverse demand curve $P = 200 - 2Q$ and has constant marginal cost $MC = 40$ with no fixed costs.

- Find the monopolist's profit-maximizing output Q_m^* , price P_m^* , and profit π . Show your derivation of marginal revenue.
- What would be the competitive equilibrium price and quantity if this market were perfectly competitive? Calculate consumer surplus under perfect competition.
- Under monopoly, calculate consumer surplus (CS_m), producer surplus (PS_m), and the dead-weight loss (DWL). Show the DWL as both the difference in total surplus and as a triangle formula.
- Calculate the Lerner Index at the monopoly equilibrium. Verify that it equals $1/|\varepsilon_D|$ using the price elasticity of demand at the monopoly point. Why does a profit-maximizing monopolist always operate on the elastic portion of its demand curve?

2. Third-Degree Price Discrimination

A museum can identify adult and student visitors and charge them different admission prices. The inverse demand curves are:

$$P_A = 100 - Q_A \quad (\text{adults}), \quad P_S = 60 - Q_S \quad (\text{students})$$

The museum's marginal cost is $MC = 20$ and fixed costs are zero.

- If the museum practices third-degree price discrimination, find the profit-maximizing price and quantity in each market and the total profit.
- If the museum must charge a single uniform price to all visitors, find the profit-maximizing price, total quantity, and profit. (Hint: derive aggregate demand by horizontally summing the two demand curves.)
- Under price discrimination, compute the Lerner Index and price elasticity of demand for each market. Verify that the inverse elasticity rule holds. Which group receives the higher markup, and why?
- Compare total welfare (consumer surplus plus profit) under price discrimination and uniform pricing. Is price discrimination welfare-improving in this case? Explain the intuition.

3. Cournot Duopoly with Asymmetric Costs

Two firms compete in quantities with inverse demand $P = 120 - Q$, where $Q = q_1 + q_2$. Firm 1 has $MC_1 = 20$ and firm 2 has $MC_2 = 40$.

- Derive each firm's best response function.
- Find the Nash equilibrium quantities, the market price, and each firm's profit.
- Compute the Lerner index for each firm. Which firm has more market power? Explain.
- How does total output and welfare compare to the symmetric case where both firms have $MC = 30$? What drives the difference?

4. Bertrand Competition

Two firms sell a homogeneous good with inverse demand $P = 100 - Q$. Firm 1 has marginal cost $MC_1 = 10$ and Firm 2 has marginal cost $MC_2 = 20$. Each firm simultaneously sets its price; the lower-priced firm captures the entire market (with equal splitting at equal prices).

- Show that $p_1 = p_2 = c$ (where c is the common marginal cost) is a Nash equilibrium when both firms have the same cost. Demonstrate that neither firm can profitably deviate.
- Show that $p_1 = p_2 = p > c$ is **not** a Nash equilibrium when both firms have the same cost. What is the intuition?
- With the asymmetric costs given above ($MC_1 = 10$, $MC_2 = 20$), find the equilibrium prices and profits. What determines the low-cost firm's equilibrium price?
- The Bertrand model predicts that two firms are enough to achieve the competitive outcome (zero profit with identical costs). Describe two real-world departures from the Bertrand assumptions that allow firms to earn positive profits.

5. Differentiated Bertrand Competition

Two firms sell differentiated products with demand:

$$q_1 = 10 - 2p_1 + p_2, \quad q_2 = 10 - 2p_2 + p_1$$

Both firms have marginal cost $c = 1$.

- Derive each firm's best response function. Are prices strategic complements or substitutes? How do you know?
- Find the Nash equilibrium prices, quantities, and profits.
- Show that the equilibrium price exceeds marginal cost. How does product differentiation resolve the Bertrand paradox?
- Suppose both firms collude and maximize joint profits, setting a common price p . Find the collusive price and each firm's profit. Compare to the Nash equilibrium. Can the collusion point be sustained as a Nash equilibrium in a one-shot game? Why or why not?

6. Stackelberg Competition

Two firms compete in quantities. Inverse demand is $P = 150 - Q$ where $Q = q_1 + q_2$, and both firms have marginal cost $c = 30$.

- a) Derive each firm's best response function and find the Cournot Nash equilibrium quantities, price, and profits.
- b) Now suppose firm 1 is the Stackelberg leader: it chooses q_1 first, and firm 2 observes this and then chooses q_2 . Using backward induction, find the Stackelberg equilibrium quantities, price, and profits for each firm.
- c) Compare the Cournot and Stackelberg outcomes from parts (a) and (b). Does the leader gain from moving first? What happens to the follower? Explain the "top dog" strategy.
- d) Compute consumer surplus and total welfare under both Cournot and Stackelberg. Which outcome is better for consumers? For society?